Thesis Portal UI: weeks 1-6 submission

Author: Nhan, Frontend & DSpace UI. **Assignment period:** 28 September-8 November 2026. **Analysis, implementation and evidence date:** 4 November 2026, Asia/Ho_Chi_Minh.

1. Purpose and scope

This package completes the frontend work assigned through week 6: research, screen inventory, component selection, wireframes, role/email flows, portal skeleton, review queues, embedded PDF viewing, decision dialogs, Editor metadata editing, Final Editor actions, approval history, and an email demonstration. Existing student submission and administration functions are retained. New collection-role administration and deposit-certificate templates belong to later weeks.

The two research targets are the existing thesis deposit portal and the standard DSpace repository UI, as agreed for this submission. The design is a project adaptation of DSpace workflow concepts, not a claim that the React interface is the DSpace Angular application or an exact visual clone.

2. Source analysis and design findings

The initial source already implemented local/optional Google login, Vietnamese/English localization, configurable submission fields, PDF upload, drafts, student tracking, assigned reviewer queues, library intake, director archival, and administration. Most interface behavior lived in a large App component. Existing documentation and September test reports lagged behind the current source, so historic checkmarks were not used as proof of current correctness.

The main gaps were an absent Editor metadata form, separate-tab PDF access, incomplete audit presentation, and a director rejection action that contradicted the week-6 return-to-Editor requirement. Backend inspection also found that the new metadata endpoint wrote abstract into description and attempted to update a nonexistent submissions.updated_at column. These were corrected without a schema migration.

The frontend retains React 18, Vite, Ant Design 5, HCMUT branding, and Vietnamese/English support. New reusable components isolate the queue, PDF preview, metadata form, history adapter/timeline, and shared review workspace. The layout provides collection context, current-stage badges, metadata/PDF columns, clear role actions, and a mobile stacked layout.

Research limitations

The repository's supplied HCMUT deposit summary was mapped to the existing form and file requirements. Its old Form 10 and Form 12 links redirected to the university home page on the research date. Current institutional policy text could not be independently confirmed. The report therefore distinguishes supplied guidance from verified current policy.

The official DSpace demo information, UI source, and DSpace 7 Configurable Workflow documentation informed the design. The current public demo is not a pinned version of this project's DSpace integration, and its authenticated screens were not visually verified. Editable annotated wireframes record the intended project design.

3. Implemented behavior

Submission and assigned queues

The existing submission form and draft/resubmission flows remain available. A shared queue wrapper supplies a consistent empty state and current-stage labels. Persistent error/retry feedback supplements existing refresh/loading behavior. Workflow annotations are now returned in student/staff lists as well as review queues, allowing actions to follow the actual configured step.

PDF preview and Editor metadata

Authorized readers can inspect the PDF inside the shared workspace. The file is fetched using the existing protected API and Authorization header; a lazily loaded PDF.js canvas renderer uses a temporary blob URL and provides previous/next page controls. Screenshot review exposed a malformed, blank legacy test fixture. It was replaced with a valid two-page PDF, and PDF.js makes page rendering and navigation explicitly verifiable. Switching files or closing the workspace aborts pending requests and releases object URLs. Open/Download and retry feedback cover unavailable preview, encrypted/invalid PDFs, or missing files.

At the active Editor step, library staff can edit Vietnamese/English titles, advisors, major, year, abstract, and keywords. The browser sends changed values; the server preserves omitted fields, validates configured core requiredness, updates the correct columns, and synchronizes the queue title. Description remains separate from abstract. A transactional metadata_updated event records the actor and before/after changes. A no-op save does not invent a change event.

Final Editor return and archive

The director interface exposes Return to Editor and Commit to Archive. Return requires a reason, moves the configured submission to the nearest previous Editor step, and preserves its academic decisions and PDF. Editor correction and acceptance bring it forward again. Director rejection to the student is disallowed by the server. Archive requires UI confirmation and remains restricted to a final director step. Configured nonfinal director steps retain acceptance.

Requests check the current step while holding the submission row lock, so stale duplicate returns cannot both commit. Frontend in-flight guards prevent duplicate decision requests. Reject/return dialogs remain open with their input when the request fails.

Approval history and email

One history adapter accepts the existing role/actorRole variants and legacy SUBMIT/RESUBMIT names. The timeline shows actors, roles, timestamps, comments, status changes, and metadata diffs. The history API restricts access to the submitter/coauthors, assigned reviewers, and authorized staff.

The existing mail service is used for local SMTP delivery. A new return notification tells library staff what to correct. Mailpit displays actual portal-generated workflow messages; it is a separate local demonstration surface. SMTP failure does not undo a committed workflow action. Email integration tests match message content to avoid confusing delayed earlier-stage mail with the decision being tested.

4. Weekly traceability and artifacts

Week	Assignment	Submission evidence
1	UI/deposit research; view inventory	research-and-requirements.md
2	Components and wireframes	Ant Design decision; wireframes.html; PNG design frames
3	Role and email flows	flows.md; use-cases.svg and editable PlantUML
4	Skeleton and task queues	Shared queue, role navigation, stage labels, feedback, responsive shell
5	PDF, decisions, Editor form	Working preview, validated metadata editing, resilient dialogs, audit diffs
6	Final Editor, history, inbox	Return/archive actions, history, local Mailpit screenshots and sanitized evidence

The dates describe assignment coverage, not retrospective completion dates. All newly collected evidence is dated 4 November 2026. Artifact navigation is in README.md; validation.md records the actual checks, and demo-script.md

provides a repeatable role-based demonstration.

5. Verification and remaining integrations

Validation includes both production builds, the existing API regression suite plus new workflow tests, actual local Mailpit integration, and browser workflows. Tests cover persistence after reload, unchanged description, actual audit payloads, stage restrictions, unauthorized readers, required reasons, repeated returns, rejection/resubmission, archive confirmation, PDF retry, failed requests, and mobile Vietnamese presentation. Exact results and reproducible commands are in validation.md.

The browser harness was corrected to prepare the disposable database before starting NestJS. The former global setup ran after the web server started and terminated its database connections. Existing browser helpers were aligned with the current read-only faculty selector and explicit English test locale.

Actual PAdES signing and certificate verification are still team dependencies. This package does not introduce a signing engine, attach a fabricated signature, or claim that archive produced a signed file. A future signing integration must provide the actual signed output and authoritative verification metadata before the UI can display success.

Live DSpace publication is not verified by this local package. The existing archive API returns dspaceDeferred=true; publication remains a separate authorized operation. Local placeholder-based tests confirm portal behavior only. The UI distinguishes archived-in-portal from publication pending/failed/confirmed.

References

• Assignment: repository docs/Week Plan (15.10 updated) - NEW PLAN.pdf, Nhan column, weeks 1-6.

• Supplied deposit guidance: repository docs/quy-dinh-nop-luu-chieu-thu-vien.md.

• DSpace official demo: https://dspace.org/demo/

• DSpace UI source: https://github.com/DSpace/dspace-angular

• DSpace 7 Configurable Workflow: https://wiki.lyrasis.org/spaces/DSDOC7x/pages/104566678/Configurable%2BWorkflow

• PDF.js examples and rendering model: https://mozilla.github.io/pdf.js/examples/

• Ant Design 5 components: https://5x.ant.design/components/overview/

See research-and-requirements.md for source status and the boundaries of each inference.

Validation and evidence

Evidence date: **4 November 2026**, Asia/Ho_Chi_Minh. Test reports contain UTC timestamps for reproducibility. Target: local Windows checkout, isolated PostgreSQL 16, local NestJS/Vite, Playwright Chromium, and Mailpit.

Results

Check	Result	Evidence
Frontend production build	Passed	Vite build; bundle-size advisory remains
Backend production build	Passed	NestJS TypeScript compilation
API regression	51 passed / 0 failed	evidence/api-report.md
Local SMTP integration	8 passed / 0 failed	evidence/email-report.md
Browser regression	8 passed / 0 failed	evidence/playwright-report.md
Production preview	Nonblank PDF; worker HTTP 200; no page errors	evidence/production-preview.json
Demonstration screenshots	Actual local UI and Mailpit capture	evidence/demo-evidence.json

What the checks establish

• Existing submission, upload, permissions, review, resubmission, portal archive, and local placeholder-publish behavior remain covered.

• Metadata persists in the correct fields; description and untouched fields remain intact; the main title matches the edited title; before/after audit data is recorded.

• Editing is restricted to an active Editor step. History rejects unrelated students and unassigned reviewers. Anonymous history requests fail.

• Return-to-Editor requires a reason, preserves academic approval and files, moves the item between the correct queues, and supports correction/forwarding/archive. Duplicate returns cannot both commit.

• Local Mailpit receives actual submission/decision/archive/return notifications; the return email contains the director's reason. SMTP failure does not invalidate the submission.

• Browser coverage includes actual nonblank PDF.js page rendering, failed metadata save, failed return and preserved reason, archive confirmation, PDF failure/retry, and Vietnamese mobile layout.

Environment and harness corrections

The test database is disposable thesis_test, on local port 5433. Suites reset it and must run sequentially. The browser runner now prepares this database before starting the API; doing so in Playwright globalSetup previously terminated an already connected API.

Existing browser helpers were updated for the read-only faculty field and explicit English test locale. Decision buttons now have stable accessible labels, including while a loading icon is present. Visual QA also exposed a malformed legacy

PDF fixture; a valid two-page fixture and PDF.js replaced the blank native-frame evidence; the browser test now checks actual rendered, nonwhite page pixels rather than only a loaded URL. Test reports are generated under frontend/docs/evidence.

Postman CLI in this environment does not support postman status. No Postman workspace, external collection, mock, or hosted service was used. Verification reuses the project's existing Jest/Supertest and Playwright suites. No simulation of a real DSpace service is claimed.

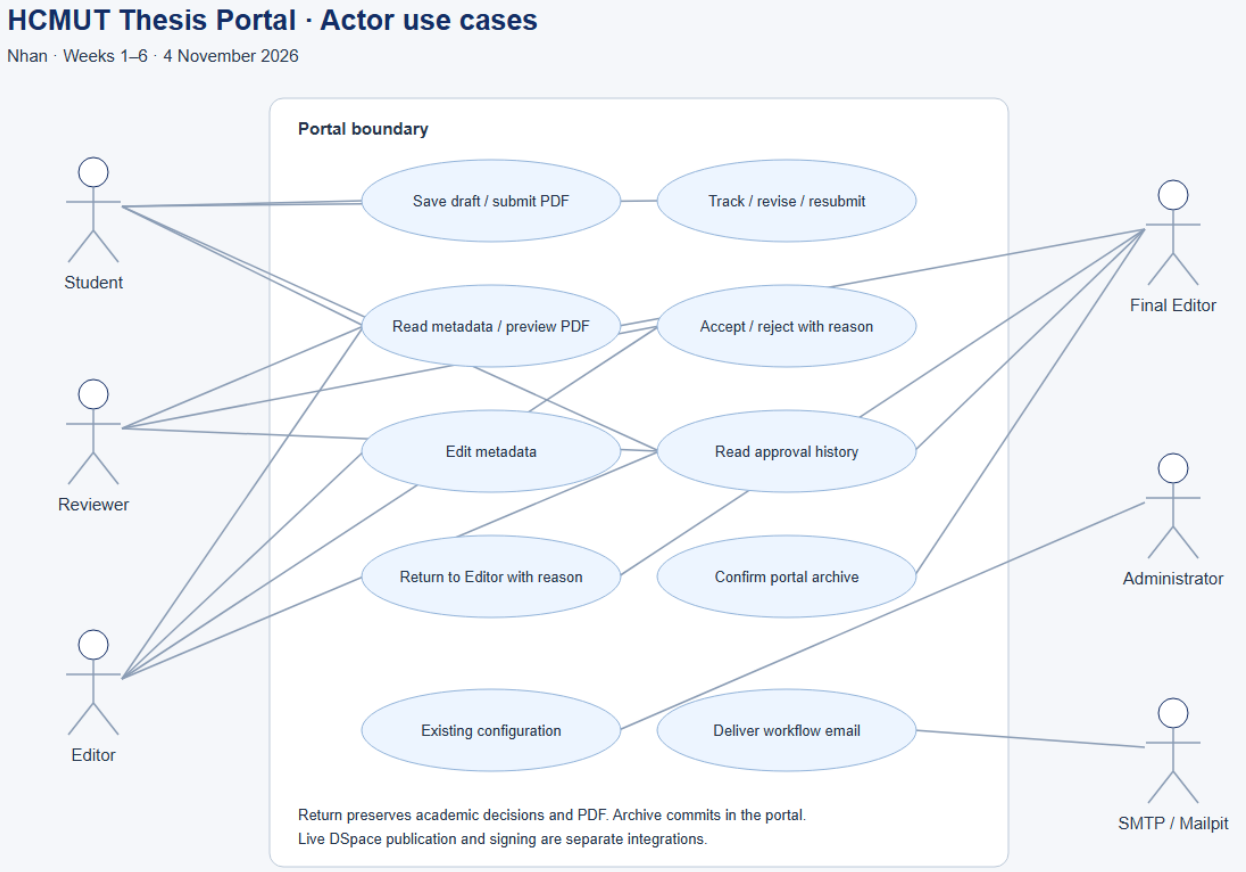
Limits and outstanding team dependencies

Integration	Status	Required before claiming completion
PAdES signing and certificate verification	Not implemented by this UI package	Real signing module, output file, and authoritative verification metadata; director archive must call it
Live DSpace publication	Not verified	Configured reachable DSpace instance, real collection, successful item/bitstream publication
Production email	Not verified	Real SMTP configuration and recipient delivery verification
Current HCMUT policy text	Not independently verified	Updated official deposit policy/form URLs; old links redirect to university home
Full keyboard/screen-reader audit	Not performed	Dedicated accessibility review beyond labels and browser interaction checks

Portal archived, a placeholder item ID, a local SMTP capture, and successful UI tests must not be presented as evidence of a verified signed document or successful live DSpace publication.

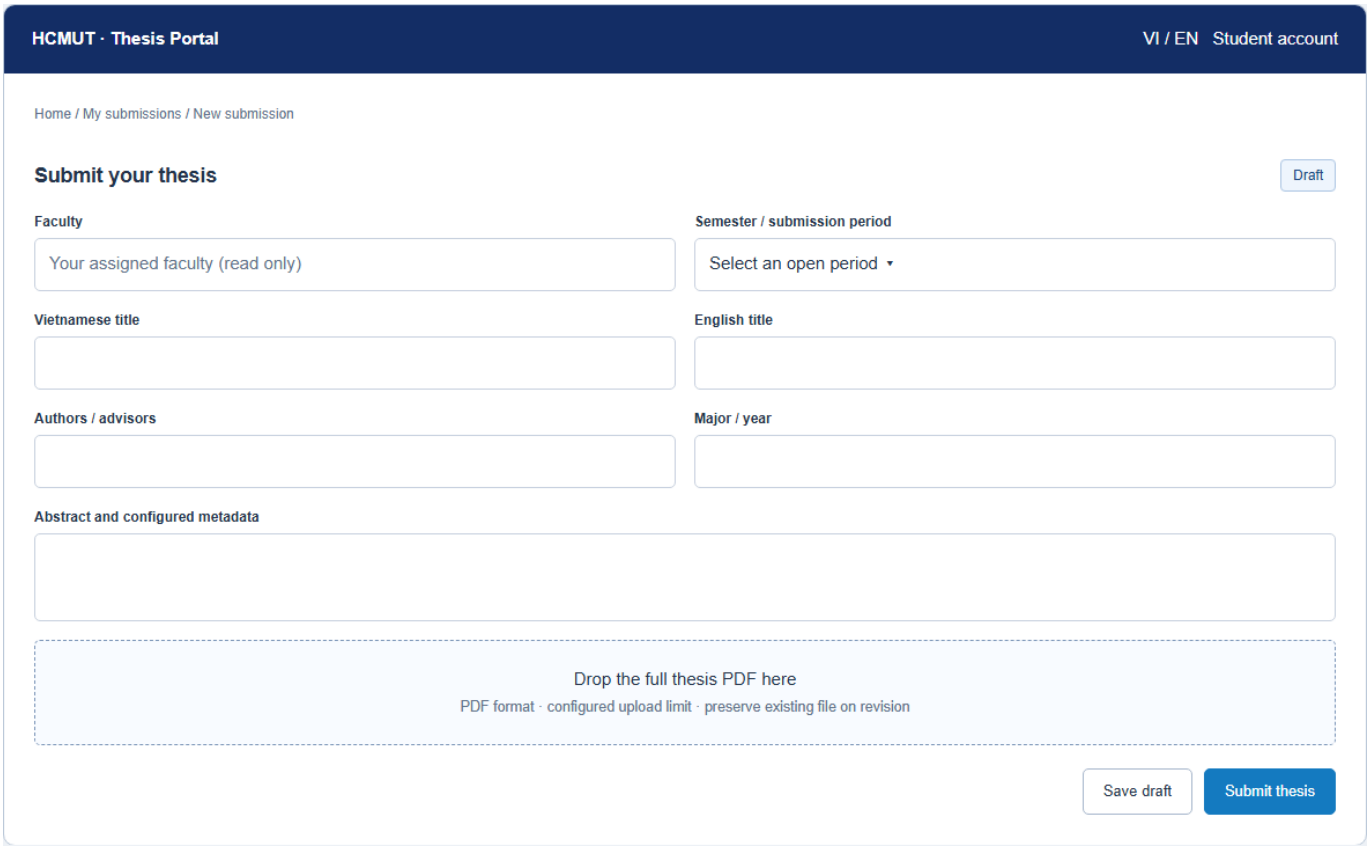
Actor use cases

Editable source: diagrams/use-cases.puml. Actors are shown only with capabilities in the current submission boundary.



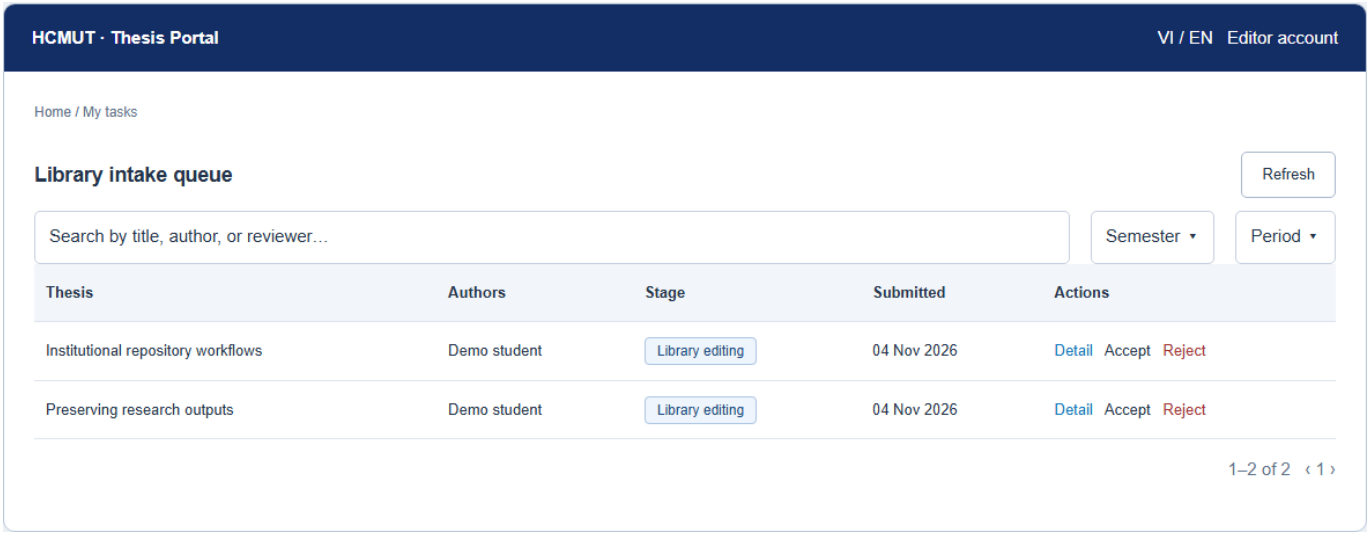
Student submission wireframe

Design frame: collection context, metadata, required PDF, and separate save/submit actions.



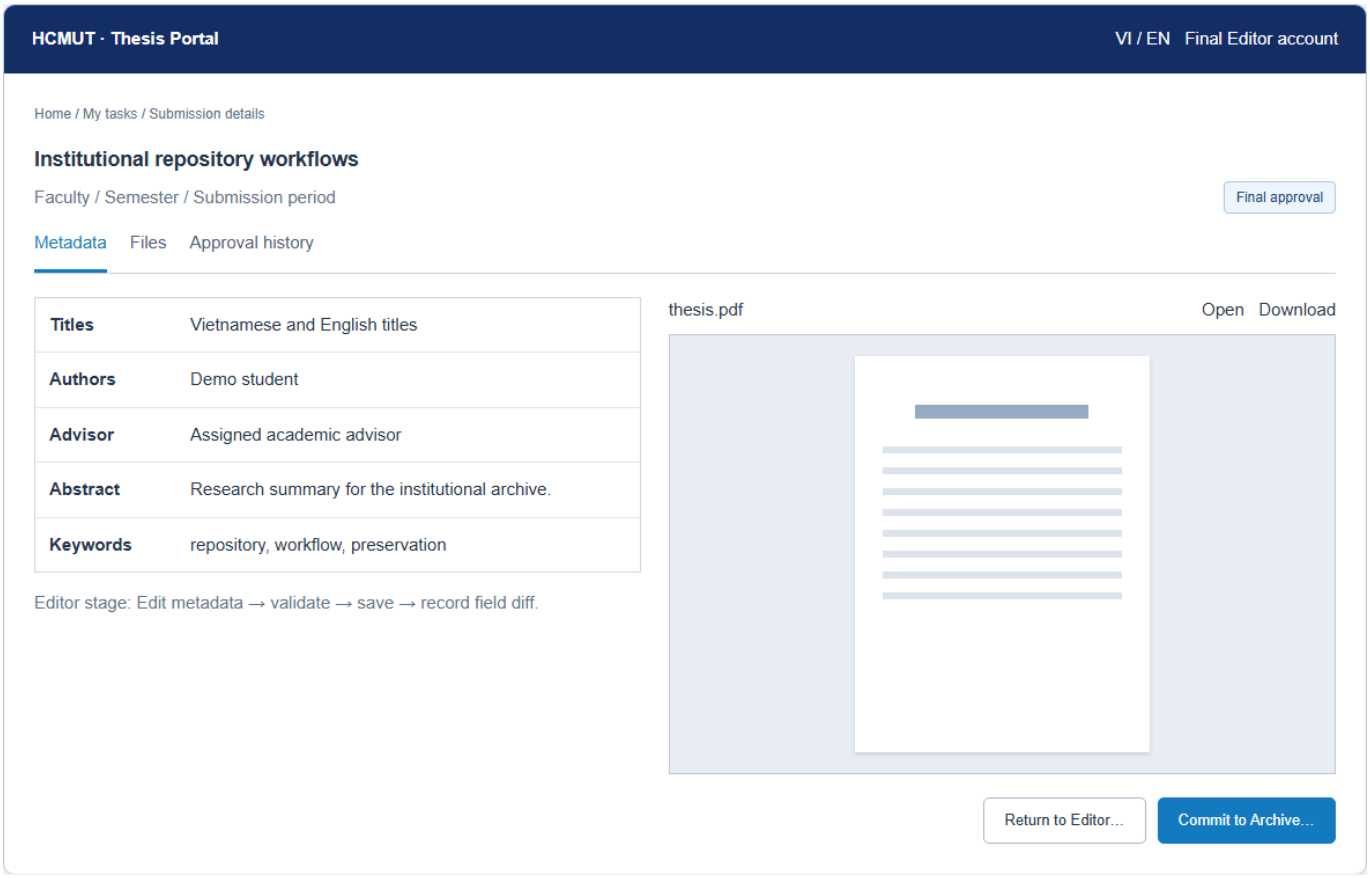
Assigned queue wireframe

Design frame: searchable assigned tasks, current stage, role actions, and feedback.



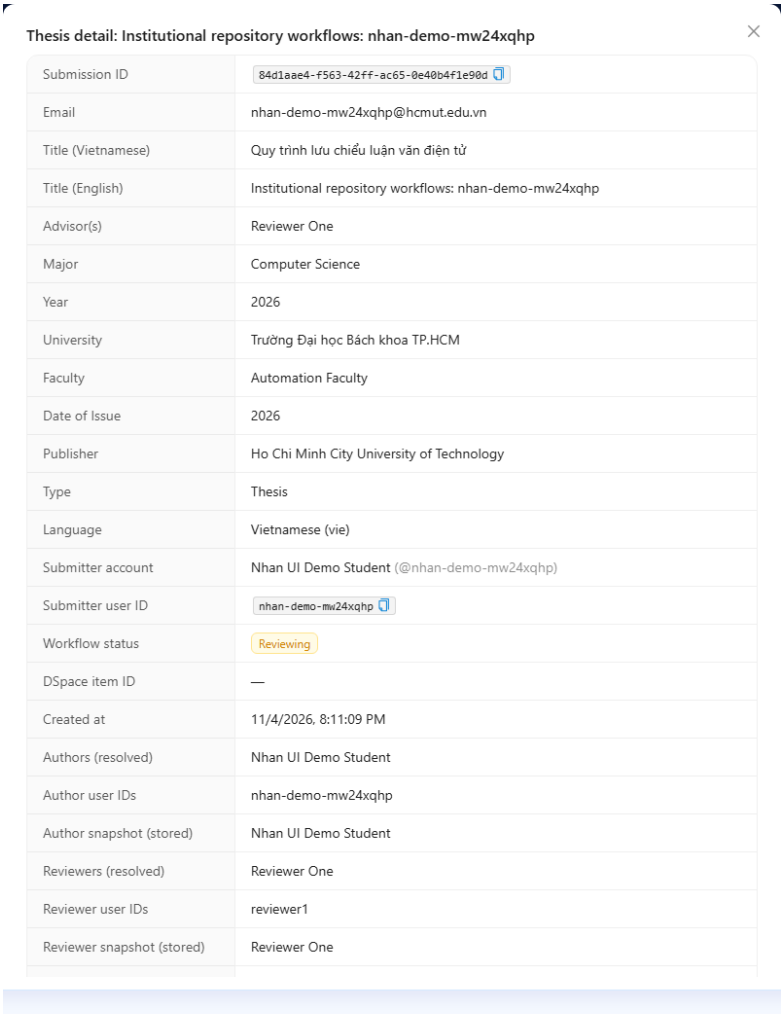
Shared workspace wireframe

Design frame: metadata/PDF columns, history, and Final Editor return/archive actions.



Before: original Editor detail

Original tracked frontend at baseline 4f7a73df rendered with current local test data; not a historical screenshot.

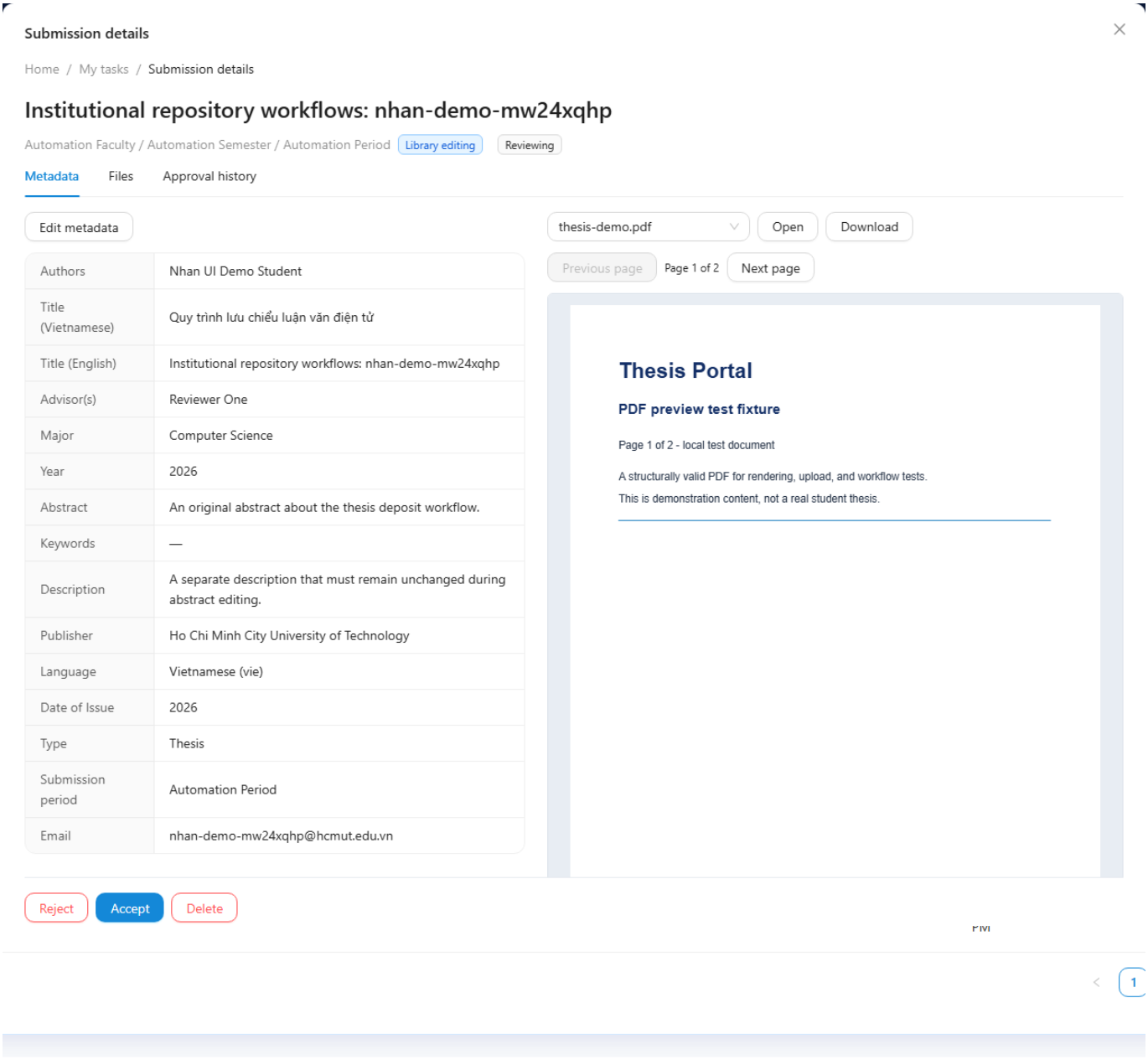


Thesis detail: Institutional repository workflows: nhan-demo-mw24xqhp

Submission ID	84d1aae4-f563-42ff-ac65-0e40b4f1e90d
Email	nhan-demo-mw24xqhp@hcmut.edu.vn
Title (Vietnamese)	Quy trình lưu chiểu luận văn điện tử
Title (English)	Institutional repository workflows: nhan-demo-mw24xqhp
Advisor(s)	Reviewer One
Major	Computer Science
Year	2026
University	Trường Đại học Bách khoa TP.HCM
Faculty	Automation Faculty
Date of Issue	2026
Publisher	Ho Chi Minh City University of Technology
Type	Thesis
Language	Vietnamese (vie)
Submitter account	Nhan UI Demo Student (@nhan-demo-mw24xqhp)
Submitter user ID	nhan-demo-mw24xqhp
Workflow status	Reviewing
DSpace item ID	—
Created at	11/4/2026, 8:11:09 PM
Authors (resolved)	Nhan UI Demo Student
Author user IDs	nhan-demo-mw24xqhp
Author snapshot (stored)	Nhan UI Demo Student
Reviewers (resolved)	Reviewer One
Reviewer user IDs	reviewer1
Reviewer snapshot (stored)	Reviewer One

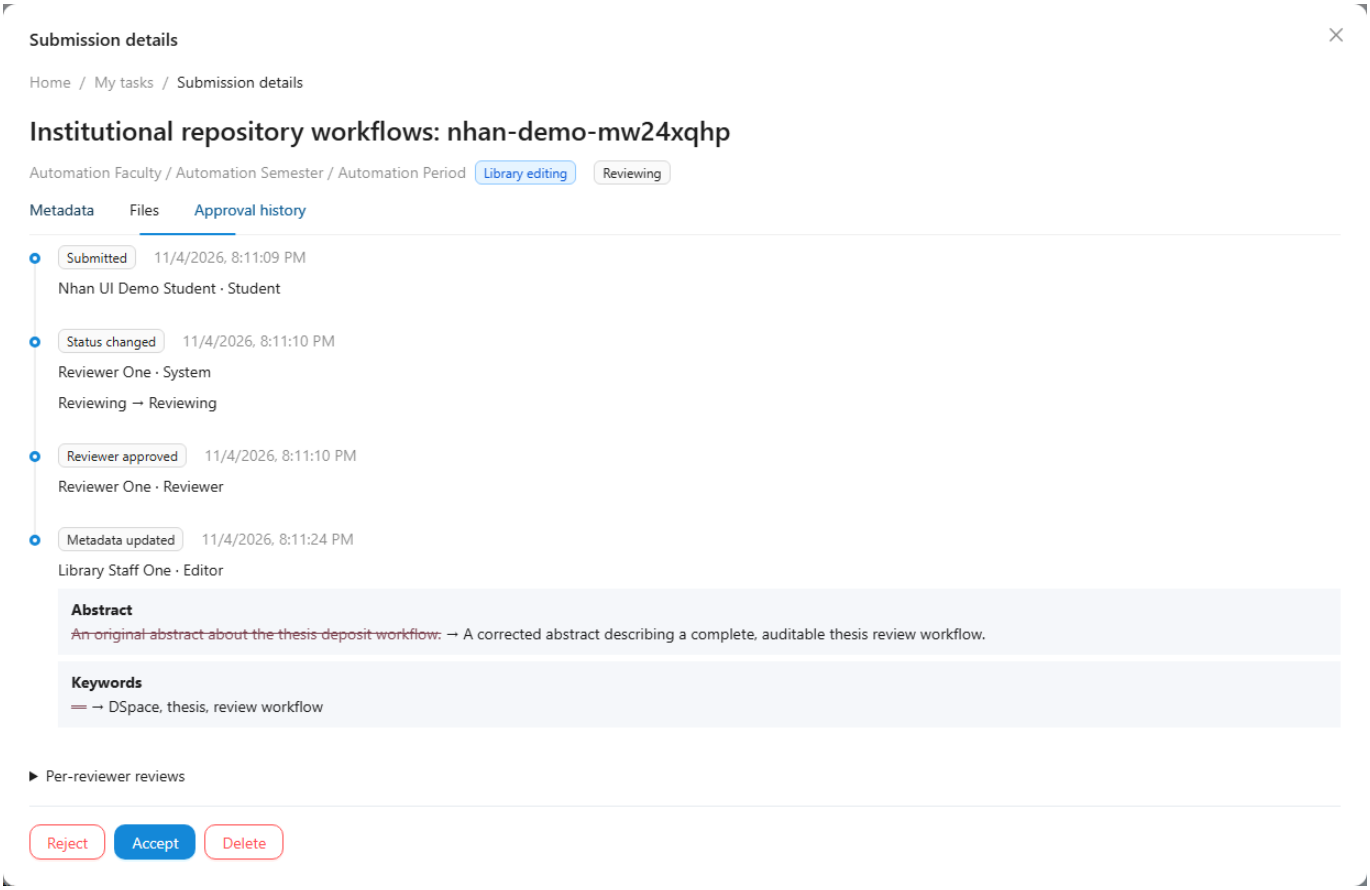
After: Editor metadata and PDF

Actual local application. Metadata and authenticated PDF preview share one workspace.



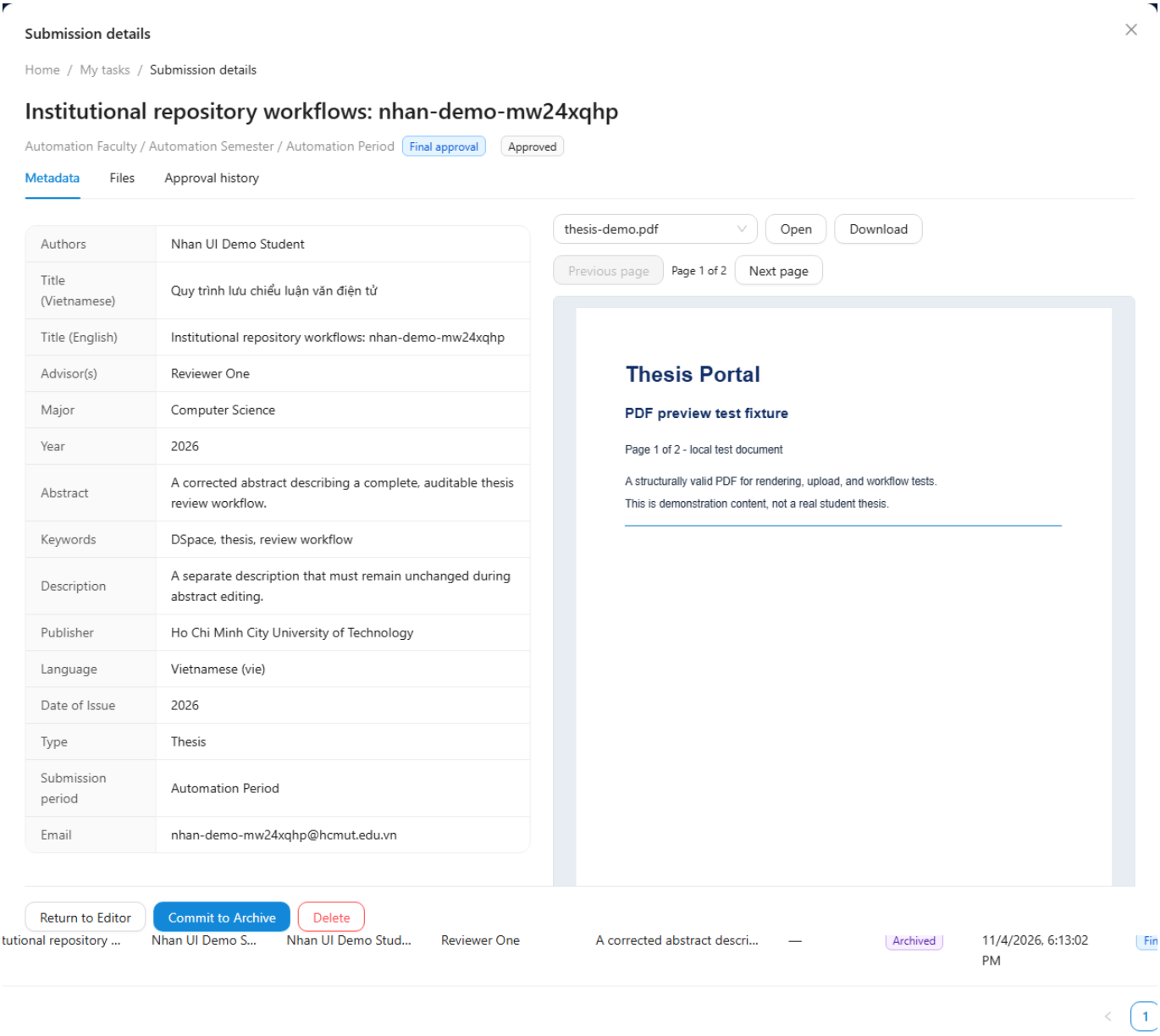
After: metadata audit history

Actual local application. The change event records the real local test actor and field-level before/after values.



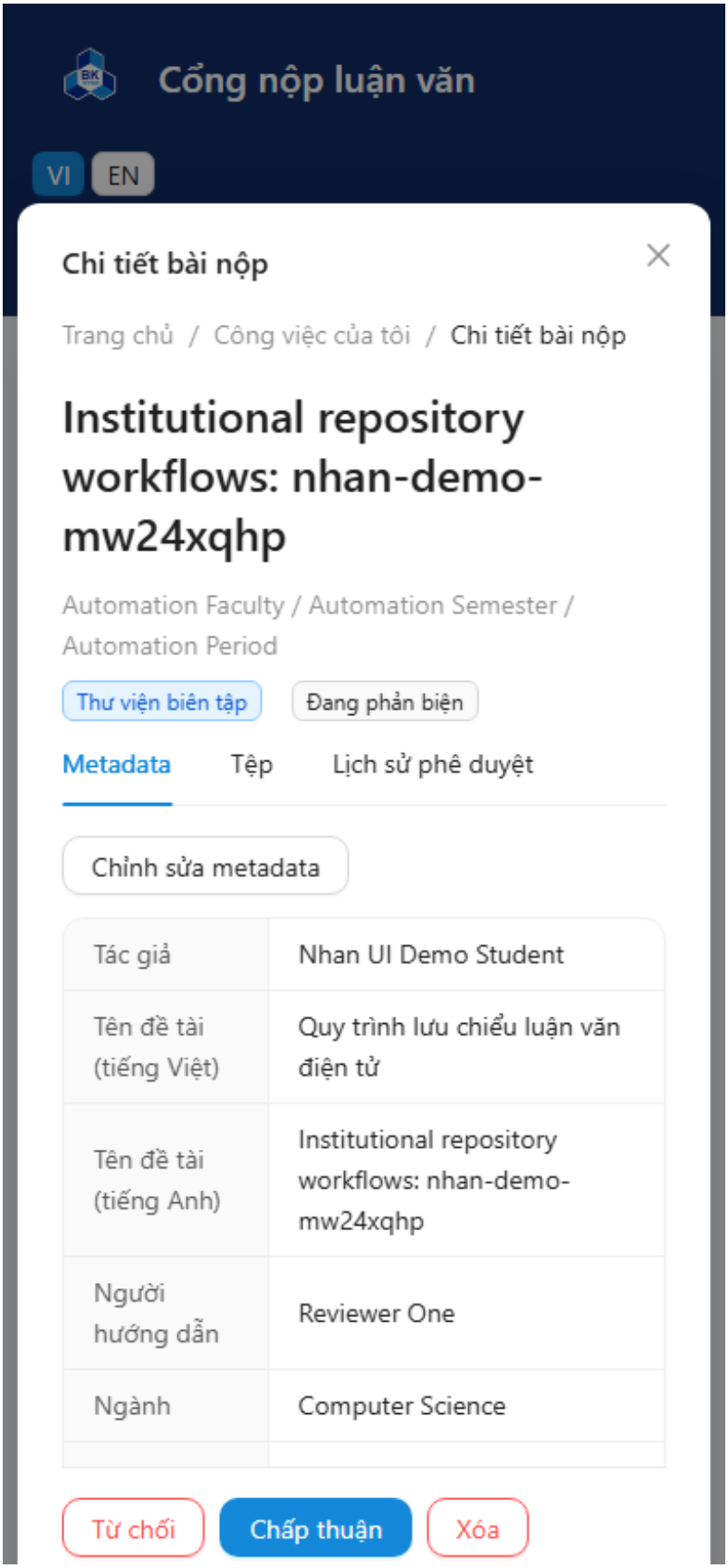
After: Final Editor

Actual local application. Return to Editor and Commit to Archive replace rejection to the student.



After: Vietnamese mobile workspace

Actual local application at 390px viewport. Content stacks vertically and supports Vietnamese.



Actual portal-generated email

Mailpit captures local SMTP messages. This is not evidence of production delivery or DSpace-generated mail.

